AI/ML Engineer Intern Assessment – Submission

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Assessment: Agentic RAG Challenge – Prerequisite & Course Planning Assistant

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1. Overview

This project implements a catalog-grounded Retrieval-Augmented Generation (RAG) system designed to assist students in course planning within an LMS environment. The system interprets academic catalog documents to answer prerequisite queries, generate course plans, and ensure all outputs are strictly grounded in source documents with verifiable citations.

2. Data Sources

The system utilizes publicly available academic catalog data, specifically:

- **20+ Course Description Pages:** Detailed prerequisites and co-requisites.
- **2+ Program Requirement Pages:** Degree rules and elective options.
- **1+ Academic Policy Page:** Grading systems and credit limits.

Each source tracks URL, Access Date, and Content Description for transparency.

3. System Architecture

The solution follows a modular RAG pipeline designed for accuracy and verification.

Pipeline Flow

```
User Input → Intake Module → Retriever → Planner (LLM) → Verifier →
Structured Output
```

Components

- **Ingestion:** HTML-based catalog scraping.
- **Chunking:** 500 tokens with 100-token overlap (optimized for prerequisite chains).
- **Embeddings:** Sentence Transformers (MiniLM).
- **Vector Store:** FAISS.
- **Retriever:** Top-k similarity search ($k=5$).
- **LLM Layer:** Structured prompt enforcing strict grounding.

4. Functional Capabilities

4.1 Grounded Answers with Citations

- All outputs strictly reference catalog content.
- Zero tolerance for unsupported claims (hallucination prevention).

4.2 Prerequisite Reasoning

- **Decision:** Eligible / Not Eligible / Need More Info.
- **Evidence:** Citation-backed reasoning.
- **Next Steps:** Required courses or necessary administrative actions.

4.3 Course Plan Generation

- Recommended course lists based on student profiles.
- Justification for each recommendation.
- Risk/assumption analysis (e.g., credit load warnings).

4.4 Clarifying Questions

- Dynamic generation of 1–5 questions if input is incomplete (e.g., missing grades or major).

4.5 Safe Abstention

- Explicitly states when information is missing from the catalog.
- Directs users to official human advisors or departments.

5. Prompt Design

The system utilizes a "System-Level Guardrail" prompt that enforces:

1. Evidence-only responses.
2. Mandatory inline citations.
3. Strict JSON or Markdown structured output.
4. Standardized refusal behavior for out-of-scope queries.

6. Output Format

Each response is partitioned into:

1. **Answer / Plan**
2. **Rationale** (Why requirements/prereqs are satisfied)
3. **Citations**
4. **Clarifying Questions**
5. **Assumptions / Missing Info**

7. Evaluation

Test Set (25+ Queries)

- Standard prerequisite checks.
- Complex multi-step chains (A requires B, B requires C).
- "Edge Case" queries (not found in catalog).

Metrics

- **Citation Coverage Rate:** Percentage of claims backed by sources.
- **Eligibility Accuracy:** Correctness of "Eligible/Not Eligible" flags.
- **Abstention Accuracy:** Correctly identifying when to say "I don't know."

8. Key Challenges & Solutions

Challenge	Solution
Inconsistent catalog wording	Used Semantic embeddings for better context matching
Multi-step prerequisites	Implemented specific Chunk Overlap strategy
Missing user data	Built an Intake logic with clarifying questions
Hallucination risk	Added a Verifier layer with strict prompts

9. Limitations

- Does not account for real-time course availability (semester-specific).
- Instructor-consent-only rules still require manual human verification.
- Database is static; real-time updates are currently not supported.

10. Future Improvements

- **Integration:** Connection with live University APIs.
- **Orchestration:** Moving to **CrewAI** multi-agent orchestration.
- **UI/UX:** Deployment via Streamlit or Gradio.
- **Advanced Logic:** Graph-based prerequisite modeling (Knowledge Graphs).

11. Conclusion

This project demonstrates a robust, production-oriented RAG system that prioritizes correctness and safety over creative freedom. It aligns with real-world academic deployment standards by ensuring every AI-generated suggestion is verifiable.